Speech Commands Unknown-Handling: Experimental Study

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SPEECH COMMANDS V0.01

KEYWORD SPOTTING

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Problem Definition

12-class keyword spotting on Speech Commands v0.01

10 Target Commands

yes no up down left right on off stop go

Unknown

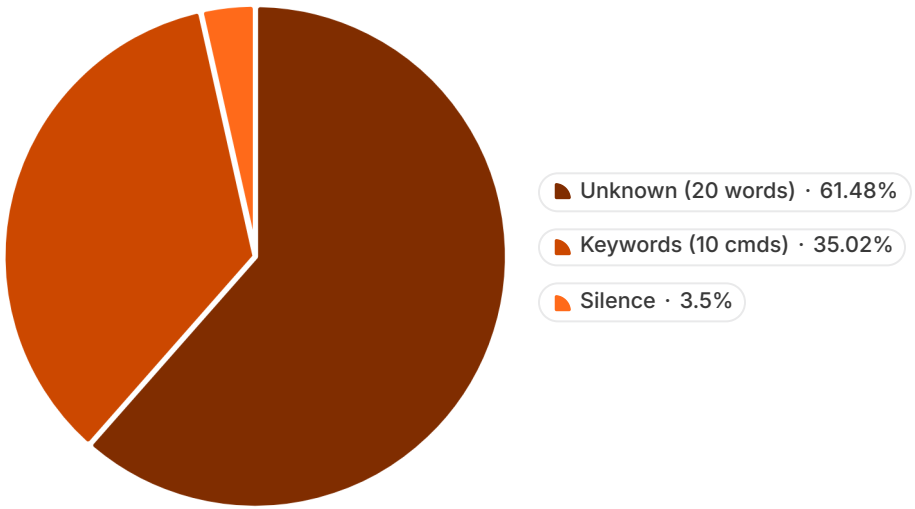
20 collapsed non-target words

Silence

Synthesized from background noise

⚠ Core challenge: **unknown** is semantically heterogeneous and heavily imbalanced

Training Set Imbalance



52,942 total training examples — **unknown accounts for 61.5%**

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Four Unknown-Handling Strategies

1

Post-Hoc Remapping

Train on all 31 labels (30 words + silence). Remap non-target predictions → `unknown` at inference. **52,942 examples/epoch.**

2

Collapsed Training

Collapse 20 non-target words → single `unknown` label before training. Class-aware balancing applied. **22,246 examples/epoch.**

3

Hierarchical Two-Head

Shared backbone with Head 1 **(silence/unknown/keyword)** and Head 2 **(10-way keyword)**. Balanced superclass batching. **~97,792 examples/epoch.**

$$\mathcal{L}_{\text{train}} = \mathcal{L}_{\text{head1}} + \lambda\, \mathbf{1}[y \in \mathcal{K}eyword]\, \mathcal{L}_{\text{head2}}, \quad \lambda = 1$$

4

Confidence-Based Rejection

Train 11-class model; unknown → uniform distribution. **22,246 examples/epoch.**

$$\hat{y}(x) = \begin{cases} \texttt{unknown}, & \text{if } \max_k p(y = k \mid x) < \tau, \\ \arg\max_k p(y = k \mid x), & \text{otherwise.} \end{cases}$$

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Model Architectures

1

BC-ResNet

Broadcasted residual learning — combines cheap 1D temporal ops with 2D frequency-aware processing. Collapses frequency axis before 1D conv; broadcasts result back. Very parameter-efficient.

2

KWT

Keyword Transformer — self-attention along the time axis of log-Mel spectrogram. Each time frame is a token. Captures long-range phonetic dependencies natively. 608K params (depth-12).

3

MatchboxNet

1D time-channel separable convolutions — depthwise 1D conv along time + pointwise channel mixing. Purely temporal processing with MFCC frontend. Lightweight and highly discriminative.

Backbone-Specific Settings

Backbone	Frontend	Optimizer / LR	Batch	Epochs
BC-ResNet-1	40-bin log-Mel	AdamW, lr=0.01	512	100
KWT-1	40-bin log-Mel	AdamW, lr=0.001	512	100
MatchboxNet-3×1×64	64-dim MFCC	NovoGrad, lr=0.05	128	100

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Experimental Protocol

The study is organized into three stages: a strategy comparison across model backbones, an augmentation grid search, and a reduced-depth KWT follow-up to probe efficiency-performance trade-offs.

1

Strategy Comparison

- Compare Post-Hoc Remapping, Collapsed Training, Hierarchical Two-Head, and Confidence-Based Rejection
- Test across **BC-ResNet, KWT, and MatchboxNet**
- Metric: validation macro-F1
- Account for sampling regime differences

2

Augmentation Grid

- 2×2×2 factorial design on Strategy A only
- Factors: **time shift, background noise, SpecAugment**
- Applied to all three backbones
- Identify the best augmentation combinations per backbone

3

Reduced-Depth KWT Follow-up

- Reduce KWT depth from **12 to 6** layers
- Vary attention mechanism: **time-only, frequency-only, both**
- Vary attention heads: **1 vs 2**
- Measure parameter efficiency and performance trade-offs

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Stage 1 Results: Strategy Comparison

Validation Macro-F1 by Backbone × Strategy

Backbone	Strat A	Strat B	Strat C	Strat D
BC-ResNet	**95.00**	91.04	94.75	93.49
KWT	**95.63**	92.55	94.78	94.25
MatchboxNet	**95.17**	92.12	94.13	94.22

Strategy B is consistently worst across all backbones.

Key Takeaways

- **Strategy A wins** for all three backbones
- **Strategy D**: competitive with ~2.5× fewer training steps than A
- **Strategy C** strong, but requires ~2× more steps than A

⚠ Sampling bias: 22K vs 53K vs 98K optimizer steps per epoch — equal epoch counts ≠ equal training exposure

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Stage 2: Augmentation Grid

2×2×2 factorial design: **time shift** × **background noise** × **SpecAugment** — Strategy A only, all backbones.

BC-ResNet

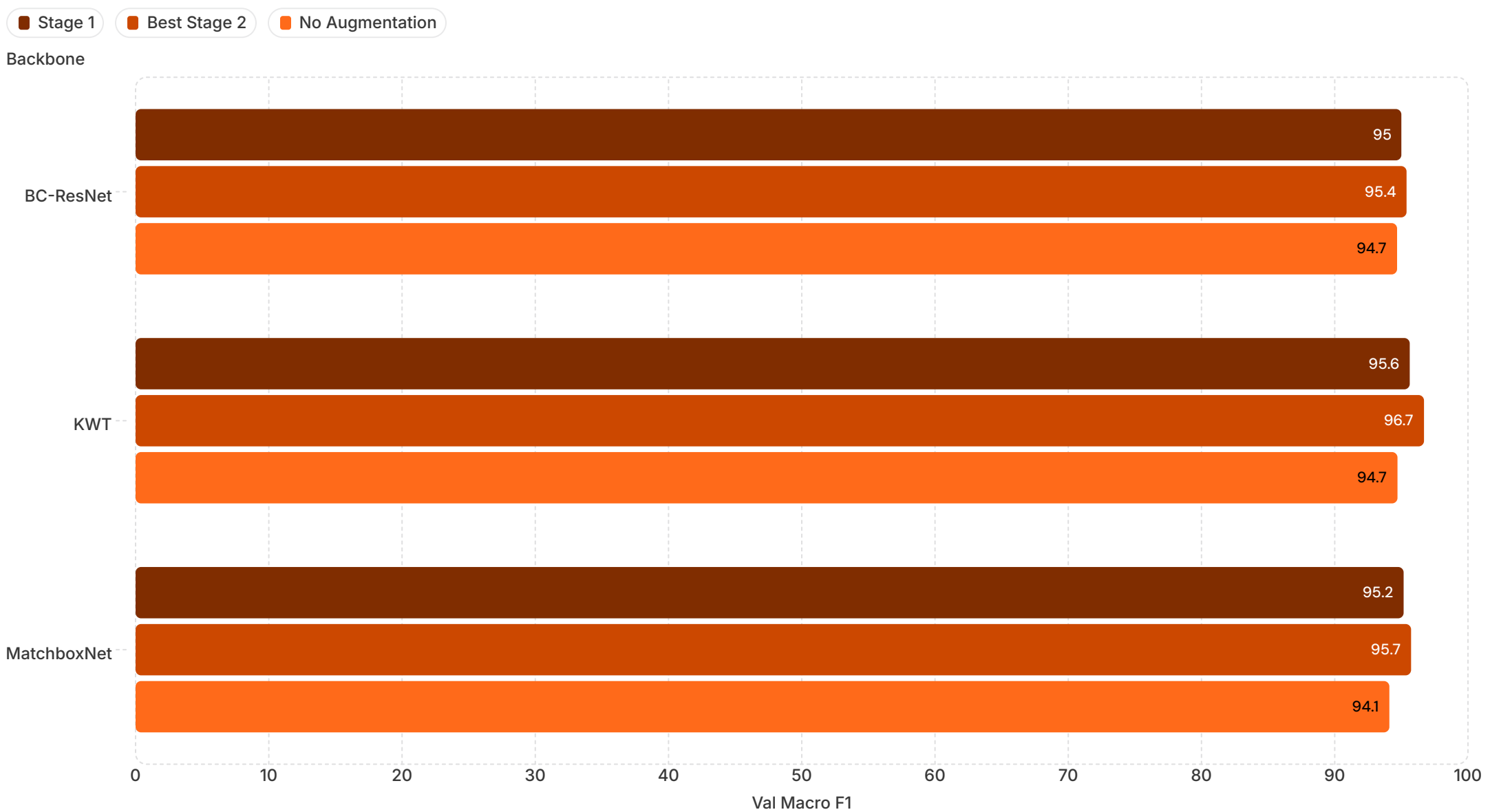
Near saturation. Best val: **aug_001** (SpecAugment, 95.39 val / 95.56 test). Minimal gain over Stage 1 baseline. Already robust.

MatchboxNet

Best run: **aug_101** (time shift + SpecAugment, 95.73 val / 95.68 test). Background noise *hurts* on average. +0.56pp over Stage 1.

KWT 🏆

Largest gain of the study. Best: **aug_111** (all three, 96.70 val / 96.45 test). **+1.9pp test macro-F1** over Stage 1. Transformer benefits most from strong regularization.



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Stage 3: Reduced-Depth KWT

Depth-6 Attention Ablation

Attention	Heads	Val F1	Test Acc %	Params
time	2	**96.44**	97.69	309K
time	1	96.13	97.66	309K
both	2	96.21	**97.79**	619K
both	1	95.98	97.49	619K
freq	2	93.38	95.64	309K
freq	1	93.55	95.64	309K

Finding

Frequency-only attention collapses — ~0.93 test F1, far below baseline

Time-only depth-6 retains strong performance at **~309K params** (2× fewer than KWT-1)

KWT paper authors' finding confirmed: **time domain > frequency domain** for attention

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Best Model: `kwt_strategy_a_aug_111`

96.70

Val Macro-F1

Best validation evidence across all 42 runs

96.45

Test Macro-F1

KWT-1, Strategy A, all augmentations enabled

97.83

Test Accuracy

Exceeds all published KWT paper benchmarks

KWT + Strategy A + aug_111 (time shift + background noise + SpecAugment) — primary recommendation

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Comparison with Published KWT Results

Paper vs. Our Runs — Test Accuracy (%)

Model	Layers	Params	Steps	Test Acc
Paper KWT-1	12	607K	23,000	97.05 ±0.23
Paper KWT-2	12	2,394K	23,000	97.36 ±0.20
Paper KWT-3	12	5,361K	23,000	97.24 ±0.24
Ours aug_111	12	608K	**10,400**	**97.83**
Ours time d6 h1	6	**309K**	**10,400**	**97.66**
Ours both d6 h2	6	619K	**10,400**	**97.79**

Efficiency Highlight

Our reduced-depth KWT uses **309K parameters** vs. **607K** for paper KWT-1, **10,400 traning steps** vs. **23,000** (equal batch size = 512), while exceeding performance.

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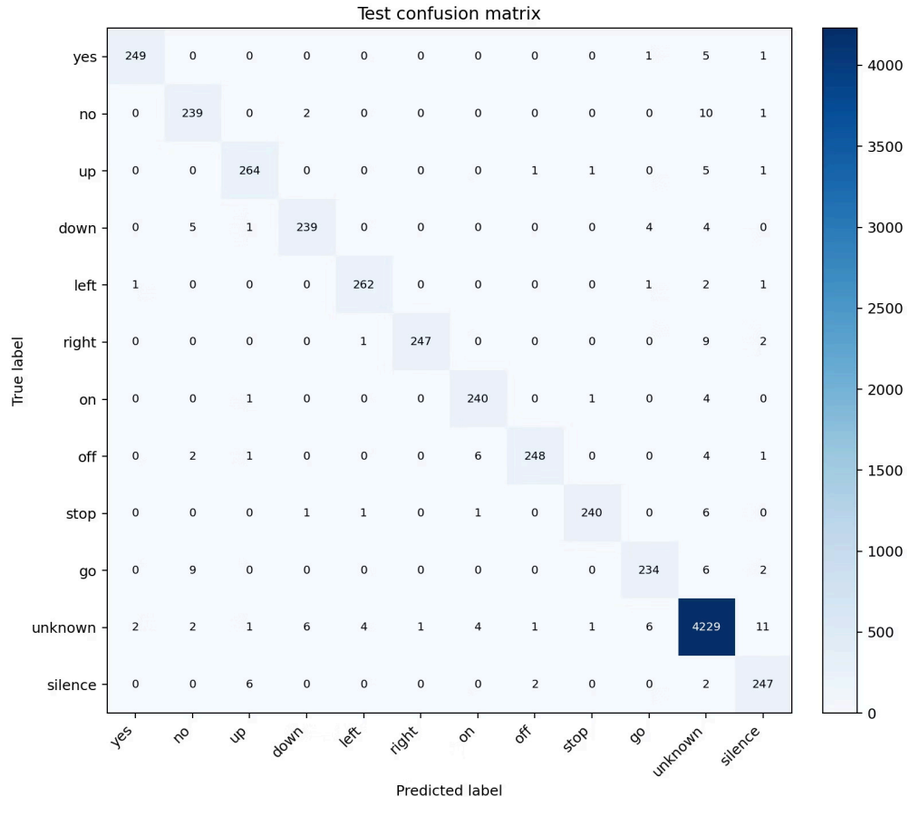
Error Analysis: Where Models Still Struggle



Remaining Errors — Best KWT Model

Confusion concentrated on a handful of acoustically similar classes:

Unknown class itself is handled well

no → unknown

Test F1 = 0.939 — shortest phoneme, easily confused

go → no

Test F1 = 0.942 — rhyming monosyllables

silence

Test F1 = 0.943 — BC-ResNet precision only 0.813; some speech absorbed into silence

right → unknown

Present across all three backbones as a weak class

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Thank You

Any Questions?

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